

RAIL-BLOC — Complete User Walkthrough (Jab Website Ready Hogi)

Poora flow ekdum start se end tak. Isko padh ke frontend design kar sakte ho — har page, har button, har click ka result.

1. Sabse Pehle: Login Page (/login)

Kya dikhega: Dark navy (#0B111E) background, center mein white card — "RAIL-BLOC Atlas Console" title, Username box, Password box, "Sign in" button. Neeche chhota text: *"SIH26027 · all data is SIMULATED"*.

Buttons:

- **Sign in** → POST /api/v1/auth/login → success pe /dashboard redirect, token clipboard/localStorage mein save
- Login fail → **red error box: "invalid credentials"**

Role ke hisaab se focus: 7 demo users hain:

User	Kya kar sakta hai
srdom_dli	Approve plans
drm_dli	Authorize plans (dusra user ho!)
controller_dli	Emergency trigger + Controller ack
engineer_dli	Manual upload + fitness certify
sm_dli	S&T signal acknowledgment
auditor	Ledger verify
admin	Solve trigger, everything

2. Home: Dashboard (/dashboard)

Kya dikhega: Top mein "Divisional Overview" heading. 4 bade KPI cards:

Card	Kya dikhata hai
Active blocks	red number = TRANSMITTED_COA + ACTIVE_GRANTED plans count
Predicted pax delay	Amber number (min) + "model estimate (B1-relative, simulated data)" label
Predicted freight detention	Blue number (min) + model estimate label
Escalated overdue	number = ESCALATED_OVERDUE demands (FSM-002 cap exhausted)

Neeche 2 sections:

- **Overdue / Escalated demands table** — jinki scheduling 3 attempts fail, human review chahiye. Red/amber rows with urgency score.
- **Machine fleet utilization** — kaunsi machine kitne jobs + travel minutes.

Neeche Plan lifecycle distribution — saare status badges (SENTINEL_PASSED, APPROVED_SR_DOM, PROVISIONAL etc.) with counts.

Auto-refresh: Har 15 sec data refresh hota hai (background polling).

Sidebar navigation (left side, hamesha visible):

Dashboard | Corridor Map | String Chart | 26-Week Calendar | Weekly Planner | Approvals | Disruptions | Audit Ledger

Role ke hisaab se kuch items chhup jaate hain (sirf APPROVALS roles ko, sirf AUDITOR ko Audit Ledger, etc.)

3. Corridor Map (/corridor-map)

Kya dikhega: Full-width MapLibre map (Google Maps jaisa dark theme). NDLS→CNB corridor ki track lines visible.

Layers:

1. Neele line = track centerline (normal)
2. Rati red = **blocked section** (live block)
3. Dashed cyan = **OHE feeding boundaries** (power isolation)

4. Moving train markers (every 10s update — RTIS mock, timetable se interpolate)

Interactions:

- **Hover section** → popup: section code, KM range, GMT index, defect count, *"SIMULATED health indices"*
- **Click section** → popup larger
- **Zoom/pan** mouse se

Buttons:

- Zoom controls (+/-) top-right
- (No action buttons — purely view + LIVE)

Important: Hamesha bottom-right **"SIMULATED DATA"** watermark, aur agar SSE stream drop ho toh top pe **"STALE DATA — actions disabled"** red banner.

4. String Chart (/string-chart)

Kya dikhega: Full-width HTML5 Canvas chart. X-axis = distance (0–250 km), Y-axis = time (00:00–23:59 = 24h window perfect).

Elements:

1. Purple diagonal lines = Vande Bharat/Rajdhani (priority 1-2)
2. Blue lines = Mail/Express (priority 3-4)
3. Amber lines = Passenger + Freight
4. Semi-transparent rectangles = **block windows** (time × km position)
5. Diagonal stripes wale boxes = **shadow bundles** (multi-department co-allocated)

Buttons/Controls:

- **< prev day / today / next day >** — day navigation
- **Scroll** = zoom (distance axis)
- **Ctrl+Scroll** = zoom (time axis)
- **Drag** = pan
- **Hover** = tooltip: jog KM + section + GMT + defects + scheduled block info

5. 26-Week Calendar (/planner/26-week)

Kya dikhega: Grid table. Rows = 12 sections (NDLS-GZB-UP, GZB-ALJN-DN, etc.), Columns = 26 weeks (W2026-08-26 se...).

Cells mein: Status badge chips (SENTINEL_PASSED, AUTHORIZED_DRM, PROVISIONAL etc.) + revision number.

Minimal buttons:

- Sirf view — no actions. Strategic 26-week horizon plans database se aate hain. Agar empty ho: message *"No strategic plans yet — trigger a STRATEGIC_26W solve from the*

weekly planner."

6. Weekly Planner (/planner/weekly) — THE BIG ONE

Kya dikhega: 2-column layout — left: plans list, right: selected plan detail.

Left side (plans list):

1. **"Trigger weekly solve"** button (top-right) — POST /optimize/solve → 202 task id → 8-20 sec baad list refresh
 - Click hote hi: "Solve queued: task_id..." message → background polling → naye plans aate hain
2. Plan rows: section code, time range, rev1 badge, **"est. pax Xm · frt Ym (model estimate)"**

Right side (Modify Parameters):

- Selected plan ka detail: status badge, content_hash (chhota), **New start / New end** datetime inputs
- **"Revise → re-enter Sentinel chain"** button:
 - POST /plans/{id}/revise → naya revision banta hai (rev2) — purana plan SUPERSEDED, naya DRAFT + sentinel_verified: false
 - Message: *"Revision created: rev 2 (...) — new revision re-enters the Sentinel chain"*

Neeche chhota text: *"FR-026: any mutation after SENTINEL_PASSED creates revision+1 at DRAFT, clears sentinel_verified and restarts the approval chain (SAFE-002)."*

Status badge update hogi: DRAFT (spinning/pulsing)

Behind the scenes (solve pipeline):

```
POST /optimize/solve
→ Redis lock (division+horizon) [agar lock busy: "a solve already
running" 409]
→ solver_runs table mein QUEUED row
→ Celery worker picks it up
→ Weather fail-closed check (stale IMD → outdoor work deferred)
→ Demands (SUBMITTED/NORMALIZED) load karta hai
→ Optional ML urgency refresh (PyTorch)
→ B1 greedy warm-start hint
→ OR-Tools CP-SAT solve (default 35s budget)
→ Sentinel 10-check validation
→ Sab pass → plans persist (SENTINEL_PASSED ya DRAFT agar GSR-2
pending)
→ machine_rosters persist
→ Ledger event + SSE publish
→ solver_runs COMPLETED
```

7. Approvals (/approvals) — SAFETY-CRITICAL SCREEN

Kya dikhega: 2-column — left: queue, right: **Action Preview Card**.

Left (queue): Plans SENTINEL_PASSED (Sr. DOM ke liye) aur APPROVED_SR_DOM (DRM ke liye). Click select karo.

Right — Action Preview Card (ye Design.md ka hero component hai):

Action Preview	[STATE BADGE]
WHAT: NDLS-GZB-UP 01:30 - 05:00 · 210 mins · rev1 hash: 4f8ab2... (chhota)	
WHY: DTT_TAMPING $\Pi \approx 0.812$ 1. TEL-001 freshness: ingested 2026-08-26 01:00Z (2h ago) [green]	
SHADOW CLUSTER: 2 co-allocated works (Track Tamping + OHE Cantilever)	
IMPACT ANALYSIS: predicted pax 55m · frt 120m [model estimate (B1-relative, simulated data)] — <i>italic label</i>	
SAFETY VERIFICATION: 10/10 CHECKS 1. G&SR-1 Absolute Block 2. G&SR-2 Interlocking Precedence 3. G&SR-3 Fail-Closed Consistency 4. G&SR-4 Power Isolation 5. G&SR-5 Headway Margin 6. MILP-C1 Section Exclusion 7. MILP-C2 Maintenance Enclosure 8. MILP-C3 Shadow Bundling 9. MILP-C4 Non-Fragmented 10. MILP-C5 Machine Conservation	
REVISION INTEGRITY: local 4f8ab2... · server 4f8ab2... decided_by [name] authorized_by: — pending—	

- | |
|---|
| <ul style="list-style-type: none"> [1. Approve & Digitally Sign] [2. Reject Plan] |
|---|

Buttons & kya hota hai:

- Approve & Digitally Sign (Sr. DOM):**
 - POST /approvals/decide {plan_id, decision: APPROVE, signature, idempotency_key}
 - Server: hash re-verify → status SENTINEL_PASSED → **APPROVED_SR_DOM**, decided_by record
 - Response: "APPROVED_SR_DOM · ledger 0x4f8ab2..."
 - Hash mismatch** → red banner: "Plan changed — reload to review latest revision. Approve disabled while the locally-held hash is stale." — button disabled
 - Same actor self-authorize (DRM)** → red: "distinct-approver violation"
- (DRM ke liye button label change)** → [Authorize & Seal] — DRM step, different label!
 - POST decide → status **AUTHORIZED_DRM**, authorized_by record
 - Footer shows: decided_by: srdm_dli · authorized_by: drm_dli
- Reject Plan** → status CANCELLED, demands CANCELLED
- Agar plan DRAFT (S&T pending acks) → yellow info box: "G&SR-2 pending acknowledgments" + "Acknowledge as Station Master" / "Acknowledge as Controller" buttons (sirf STATION_MASTER/CONTROLLER roles ko)

Special case — S&T plan (G&SR-2):

```
DRAFT (signals pending)
→ SM acknowledge → controller acknowledge
→ plan → SENTINEL_PASSED (hash sealed)
→ phir approve → authorize flow
```

8. Disruptions (/disruptions) — EMERGENCY CONSOLE

Kya dikhega: Left: incident log form + list. Right: semantic notes card.

Button flow:

- Preview blast radius** button:
 - Section ID select (12 sections)
 - Type select (TRACK_FRACTURE, OHE_BREAKDOWN, SIGNAL_FAILURE, OTHER)
 - Duration input (15-1440 min)
 - Click → **Blast Radius Modal** khulta hai:

1. Emergency Confirmation Modal – blast radius

- Trains currently held: 3
 - Plans that will be superseded: 2
 - Affected sections (incl. adjacent via feeding map): 4
- [] I acknowledge the blast radius above (API-001 modal).
- Advisory revocations will be issued by the Emergency Service;
- the re-plan is PROVISIONAL until I acknowledge it.

[2. Fire emergency breakdown]

- Checkbox ke bina "Fire" button disabled — **pehle confirmation chahiye** (API-001)

2. Fire emergency breakdown:

- POST /emergency/breakdown → 201
- System kya karta hai (behind the scenes):
 1. Section pe active/planned block check (pass/fail)
 2. Incident create (adjacent sections check → coalesce)
 3. Advisory revocation → TRANSMITTED_COA/ACTIVE_GRANTED plans → SUPERSEDED_EMERGENCY
 4. Displaced demands → SUBMITTED
 5. Corridor-scoped CP-SAT re-plan (≤35s budget)
 6. Sentinel structural checks (synchronous, sub-second)
 7. Plan persist → PROVISIONAL (incident bound) + ledger + SSE
 8. Response + measured wall time
- Response message: "✓ PROVISIONAL plan 8f2ab1... created in 12.3s (NFR-002 budget incl. synchronous structural re-check). Coalesced: no. Superseded: 2 plan(s). Awaiting Controller acknowledgment."

3. PROVISIONAL plan item → purple badge + "Acknowledge as Controller" button:

- POST /emergency/incidents/{id}/acknowledge
- → incident ack → outbox transmit allowed → COA bridge ack → TRANSMITTED_COA
- Controller ke ack ke bina plan ka kuch nahi hota (deadlock gate — by design)

List mein incidents: Section, type, est duration, coalesced tag, provisional plan id, controller ack status.

9. Audit Ledger (/audit-ledger)

Kya dikhega: Full-width table (sirf AUDITOR/ADMIN).

seq	event	actor	prev_seq	prev_hash →hash	payload	at
12	PLAN_SENTINEL_PASSED	worker	11	4f8ab...→ 0x...	{"plan_id": ...}	...

Buttons:

1. Verify chain (REPEATABLE READ):

- GET /ledger/verify → result card:

1. tamper-EVIDENT chain intact – verified 54/54

Method: full re-hash from sequence 1 under REPEATABLE READ snapshot isolation (FR-023).

- Broken chain → red "X CHAIN BROKEN"

2. Search input (event_type filter)

3. Infinite scroll/pagination

Neeche chhota text: "tamper-evident, not tamper-proof — Rules.md §3"

10. COMPLETE E2E DEMO FLOW (Demo script ke liye — 5 min)

Time	Kya karte ho	Kya hota hai system mein
0:00	Login srdom_dli	JWT mint hua
0:20	Dashboard	KPIs + escalated table
0:40	Weekly Planner → Trigger solve	CP-SAT solve, 8-20s
1:00	Plans list refresh	SENTINEL_PASSED plans dikhte hain
1:15	Approvals → plan select	Action Preview Card: 10/10 checks

Time	Kya karte ho	Kya hota hai system mein
1:30	Approve (Sr. DOM)	APPROVED_SR_DOM + decided_by
1:45	Login drm_dli → Authorize & Seal	AUTHORIZED_DRM + authorized_by (distinct!)
2:00	Login controller_dli → Plans → plan info	Plan AUTHORIZED_DRM
2:15	Transmit button	T-2h structural re-check → outbox → COA ack → TRANSMITTED_COA
2:30	Activate → Complete-fitness → Archive	ACTIVE_GRANTED → COMPLETED_FITNESS → ARCHIVED_SEALED
3:00	Disruptions → Preview → Fire breakdown	Blast radius modal → PROVISIONAL plan
3:20	Acknowledge as Controller	PROVISIONAL → transmit → TRANSMITTED_COA
3:40	Login auditor → Verify chain	54/54 unbroken
4:00	Modify-after-verify attempt (pehle wala plan revise karo)	rev2 → DRAFT; approve kiro → 409 HASH_MISMATCH
4:30	SIMULATED DATA watermark dikhao	Honesty label
5:00	Summary: "ML estimates; CP-SAT decides; Sentinel verifies; humans authorize; COA executes."	Wrap-up

11. HARR BHI KYA HAIN — STATUS BADGES (har jagah same)

Badge	Color	Matlab
DRAFT	Gray	Abhi banaya, Sentinel pending
SENTINEL_PASSED	Green	10/10 checks pass, approval ready
APPROVED_SR_DOM	Blue	Sr. DOM ne decide kiya
AUTHORIZED_DRM	Amber	DRM ne authorize kiya
TRANSMITTED_COA	Blue	COA ko mila (ack hua)
ACTIVE_GRANTED	Red	Line isolated, work chalu
COMPLETED_FITNESS	Green	Khuli line
ARCHIVED_SEALED	Gray	History sealed
SUPERSEDED	Gray	Revision replace hua
SUPERSEDED_EMERGENCY	Amber	Emergency ne replace kiya
CANCELLED	Gray	Reject/cancel
FAILED_ESCALATE	Red	3 attempts fail — human review

Badge	Color	Matlab
PROVISIONAL	Purple	Emergency plan, Controller ack pending

Har badge automatically 3 cheezein dikhata hai: icon + text + color (WCAG 1.4.1 colorblind-safe).

12. IMPORTANT DESIGN POINTS

1. **STALE DATA overlay:** SSE stream drop hote hi (Redis down/crash) → **sab action buttons DISABLED** + persistent top banner. Controller kabhi bhi stale data pe action nahi kar sakta.
2. **SIMULATED DATA watermark:** Hamesha bottom-right, non-dismissible (har synthetic layer pe).
3. **Model estimates label:** Har delay figure ke saath *"model estimate (B1-relative, simulated data)"* — fake fact kabhi nahi.
4. **Hash mismatch banner:** Approve button disabled agar local hash $\neq$ server hash.
5. **Distinct approver:** DRM button label ALAG ("Authorize & Seal") + footer mein decided_by/authorized_by display.
6. **Emergency modal:** BLAST RADIUS dikhana mandatory — confirmation checkbox ke bina fire nahi ho sakta.

13. FRONTEND ARCHITECTURE (design ke liye)

- **Pages (9):** Login, Dashboard, CorridorMap, StringChart, Planner26Week, PlannerWeekly, Approvals, Disruptions, AuditLedger
- **Shared components:** NavDrawer (role-filtered), StatusBadge, Card, ActionButton (stale pe disabled), PreviewCard, StaleOverlay, SimulatedWatermark
- **Live events (SSE):** BLOCK_ACTIVATED, PLAN_REVISED, SIGNAL_ACK, BLOCK_TRANSMITTED, PROVISIONAL_PLAN_CREATED, SOLVE_COMPLETED, SOLVE_FAILED — inko live badges/markers update karne ke liye use kar sakte ho
- **Auth:** localStorage token, JWT claims role/division — nav + permissions isi se
- **Strict TypeScript** — tsc --noEmit build gate

Ab jo bhi design karna hain — login card, dashboard grid, map, string chart, 10-check card, emergency modal, ledger table, badges — sab ka exact behavior upar hai. Isi hisaab se polish kar do, aur frontend-ready. Koi aur cheez chahiye toh bolo!